

Supplementary Materials

Explainable Artificial Intelligence Assisted Modeling of Malachite Green Adsorption onto SBA-15-Zn-Fe Composite

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Supplementary Table S1. Independent internal test-set performance of the prespecified multilayer perceptron (MLP) and alternative machine-learning models using the fixed 80/20 training-test partition.

Model	Model configuration	Test R^2	Standardized-response MSE
Multilayer Perceptron (MLP)	Hidden layers: 100-50-25-12; activation: tanh; solver: L-BFGS; L2 regularization: 0.0001; maximum iterations: 2000; random_state: 42	0.9624	0.0449
Random Forest	Estimators: 100; maximum depth: unrestricted; max_features: 1.0; bootstrap: True; random_state: 42	0.9872	0.0153
Gradient Boosting	Estimators: 100; learning rate: 0.1; maximum depth: 3; loss: squared error; random_state: 42	0.9804	0.0234
Support Vector Regression (SVR)	Kernel: radial basis function; C: 1.0; epsilon: 0.1; gamma: scale	0.6171	0.4574

Note: The complete dataset ($n = 315$) was divided once into a training subset ($n = 252$) and a strictly held-out internal test subset ($n = 63$) using random_state = 42. Input and response standardization parameters were estimated using the training subset only and then applied to the test subset. MSE values are reported in standardized-response units. The alternative models were evaluated using the scikit-learn configurations shown above, without cross-validation-based hyperparameter optimization. Bold values indicate the best independent test-set performance.

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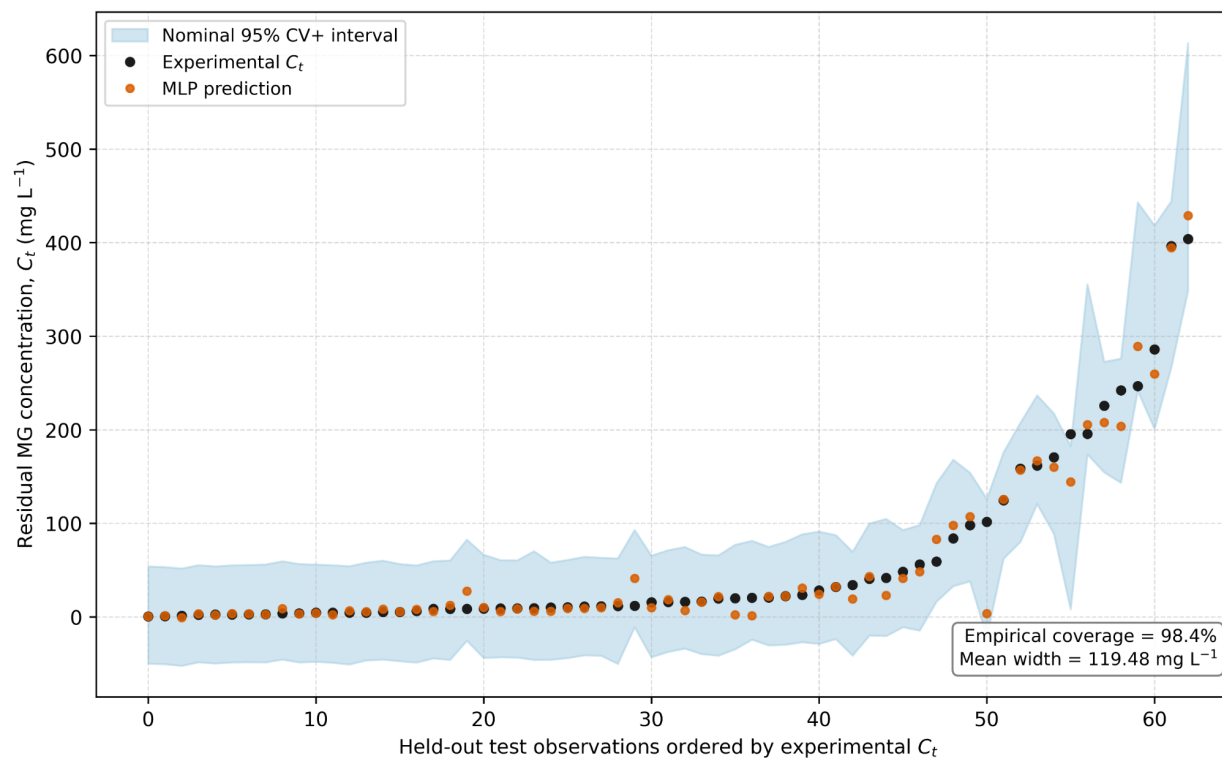
Supplementary Table S2. Textural parameters of SBA-15 and the SBA-15-Zn-Fe composite obtained from nitrogen adsorption analysis.

Material	S_{BET} (m^2g^{-1})	V_{T} (cm^3g^{-1})	d_{BJH} (nm)
SBA-15	734.79	0.6370	3.93
SBA-15-Zn-Fe composite	102.12	0.0925	4.61

Note: S_{BET} , BET specific surface area; V_{T} , total pore volume; d_{BJH} , BJH average pore diameter. The values were obtained from the available nitrogen adsorption analysis records. These textural parameters do not independently quantify the Zn and Fe contents or confirm retention of the ordered hexagonal SBA-15 structure.

Interpretive scope. Relative to SBA-15, the modified composite showed lower BET specific surface area and total pore volume and a higher BJH average pore diameter. These differences demonstrate a marked alteration of the measured textural properties after the modification procedure. Nitrogen adsorption alone, however, cannot determine whether Zn- and Fe-containing species are located inside the pores, deposited on the external surface, or associated with structural alteration.

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Supplementary Figure S1. Five-fold CV+ prediction intervals for the strictly held-out internal test subset ($n_{te} = 63$). Observations are ordered by their experimentally measured residual MG concentrations, C_t , solely for graphical presentation. Black circles represent experimental values, orange circles represent MLP point predictions, and the shaded region represents nominal 95% CV+ prediction intervals constructed from out-of-fold absolute residuals obtained exclusively within the development subset. The interval boundaries for adjacent ordered observations are connected only for visual clarity. Empirical test-set coverage was 98.41% (62 of 63 observations), and the mean and median interval widths were 119.48 and 108.04 mg L⁻¹, respectively. The intervals are unconstrained; lower bounds below zero are statistical limits and should not be interpreted as physically possible concentrations. Test responses were used only to order the observations and evaluate coverage, not to construct or calibrate the intervals.